

Executive Summary

C-suite data and governance executive with a 15-year track record of building and leading large-scale health data infrastructures that serve researchers, advance scientific discovery, and maintain public trust. Currently Chief Data Officer at Sage Bionetworks, managing a 3.8+PB multi-omics data platform and a \$40M+ research data portfolio spanning genomic, transcriptomic, proteomic, and metabolomic datasets. Previously established the UK's national Trusted Research Environment network at HDR UK, connecting 100+ NHS organisations and data custodians to serve 45,000+ researchers, and contributed to the management of EMBL-EBI's 400+PB genomic data estate.

Deep expertise in data governance as a strategic enabler of research impact. At the ICO, provided direct strategic guidance to **Our Future Health**, NHS England's Federated Data Platform, and Trusted Research Environments on data protection, consent frameworks, and responsible use of participant data. Co-authored national guidance on AI and data protection and published foundational work on TRE best practices, GA4GH genomic data sharing standards, and FAIR data principles. Champion of data quality, trustworthiness, and transparency through innovations such as Data Use Registers and the Data Use Ontology.

Proven ability to build and scale high-performing, multidisciplinary teams (50–200+ people) across data engineering, data operations, analytics, and architecture. Experienced in developing researcher-facing data products and platforms that provide intuitive, secure access to complex biomedical datasets. Combines strategic vision with operational rigour and is equally effective in presenting to boards and regulators as leading technical architecture decisions. Deep technical foundation spanning cloud infrastructure (AWS, Google Cloud, Azure), AI/ML systems (PyTorch, TensorFlow), genomic databases, and healthcare data platforms (OMOP, REDCap, TransMART, Synapse), with fluency in both Agile delivery and enterprise governance frameworks.

Professional Experience

Sage Bionetworks

(REMOTE) SEATTLE, UNITED STATES OF AMERICA

Chief Scientific & Data Officer - Sage Bionetworks

March '24 –

- **Portfolio & Financial Management:** Direct accountability for a \$40M+ annual project portfolio spanning 50+ concurrent initiatives across federal agencies (NIH: NIA, NCI, NINDS, FNIH, NIMH, NHLBI, NCATS, ARPA-H), foundations (Gates Foundation, AACR, Children's Tumor Foundation, N-TAP, RWJF, Helmsley Trust, GFF), and commercial partnerships (pharmaceutical and industry contracts with AACR, Wellcome Leap, DiME). Manage diversified funding mix demonstrating sophisticated financial planning and resource allocation across multiple funding cycles, positioning data as a core organisational asset driving both scientific impact and commercial value. Lead commercial contract negotiations with industry partners, establishing sustainable revenue streams and partnership models that balance mission-driven research with financial sustainability.

- **Leadership & Team Management:** Lead Sage's largest team of 50+ professionals (Science & Data), comprising director-level scientists, clinical-bioinformatics experts, software developers, and technical project managers across seven specialised sub-teams – a structure comparable to leading heads of data engineering, data operations, analytics, and architecture. Developed performance frameworks and career progression pathways, achieving strong retention in a competitive market. Built and scaled a multi-petabyte genomic data infrastructure (~3.7PB) for storage, curation, and researcher access to biomedical datasets including genomic, transcriptomic, proteomic, and metabolomic data, ensuring FAIR compliance and AI-readiness through standardised machine-readable metadata.

- **Strategic Partnerships & Data Products:** Established strategic partnerships with major technology providers, biobanks, and research institutions to enhance data infrastructure and expand researcher-facing data products and services. Led development of **Synapse knowledge portals** – data-centric products serving thousands of scientists and researchers globally. Secured new funding through competitive grant applications and industry partnerships. Maintained high external visibility through conferences, international committees, and publications, positioning Sage as a leader in trusted biomedical data infrastructure.

- **Data Governance & Quality:** Implemented robust data governance frameworks and standardised operating procedures across Sage's multi-programme portfolio, championing data quality as a strategic enabler of research impact. Established data curation standards and quality assurance processes for biobank-scale datasets serving the global research community. Delivered trusted data infrastructure underpinning major national programmes in Cancer (HTAN, AACR GENIE, MC2), Alzheimer's and

Exceptional Longevity (AMP-AD, ELITE), Rare Diseases (NF, AMP-ALS), and Digital Health, enabling researchers to access curated, analysis-ready datasets through Synapse knowledge portals.

Information Commissioner's Office

LONDON, UNITED KINGDOM

Head of AI & Data Science - Technology, Innovation & Enterprise Directorate, ICO May 22 – Feb 24

- **Strategic Leadership:** Founded and scaled the AI & Data Science Technology Policy team from zero to a national hub for regulatory policy development, with direct influence on UK AI governance and health data regulation. Built a data-driven culture within the directorate, embedding evidence-based decision-making using data insights and analytics. Pioneered strategies that balanced technological advancement with data privacy, research ethics, and ethical AI use, setting new standards in regulatory frameworks for health and research data.
- **Policy Innovation & Data Governance:** Co-authored ICO's Guidance on AI and Data Protection, the AI Risk Assessment Toolkit, and frameworks for responsible data governance in health research, adopted by organisations across public and private sectors. Provided direct strategic guidance to Our Future Health on data protection, consent frameworks, and participant data governance, alongside NHS England's Federated Data Platform, Great Ormond Street Hospital DRIVE, and Trusted Research Environments (TREs) – shaping governance approaches for programmes collectively managing data from millions of participants. Championed governance as a strategic enabler of innovation rather than an impediment, developing practical frameworks that built public trust while unlocking research value.
- **Cross-Sector & Executive Collaboration:** Engaged with government bodies including DSIT, DHSC, and NHS England to shape AI legislation and health data policy, fostering collaborative approaches across regulatory agencies and international Data Protection Authorities. Worked cross-functionally with senior leaders across technology, legal, policy, and communications teams. Built relationships with healthcare data custodians, NHS organisations, and industry partners to champion Privacy by Design and data protection by default, providing practical guidance that balanced innovation with regulatory compliance and public trust.

Health Data Research UK, DARE UK, Visiting Scholar EMBL-EBI

LONDON, UNITED KINGDOM

Chief Technology Officer & Dir. of Engg - HDR UK & Technical Lead - DARE UK Jan 19 – Jul 22

- **Portfolio & Financial Oversight:** Managed £69M+ portfolio of Health Data Research infrastructure projects, with direct accountability for budget allocation, milestone delivery, and financial reporting to funders. Oversaw multi-year programme budgets, ensuring efficient resource utilisation and demonstrating clear return on investment to government and charitable funders. Led commercial strategy and product economics for the HDR Innovation Gateway, developing sustainable operational models and demonstrating return on investment to government and charitable funders through clear impact metrics.
- **National Leadership & Researcher Services:** Built and led teams of 200+ in-house and external specialists, establishing a user-centric service delivery model focused on researcher experience. Established the national Trusted Research Environments (TREs/SDE) network, creating federated infrastructure connecting 100+ NHS organisations, biobanks, and data custodians, serving 45,000+ researchers with secure access to linked health records, genomic data, and cohort study datasets. Championed data quality standards and FAIR principles across the network, achieving consistent high performance against SLAs and KPIs while maintaining GDPR compliance and public trust.
- **Strategic Development & Ecosystem Partnerships:** Led multi-vendor digital transformation through the Health Data Research Innovation Gateway – a data discovery and access platform enabling researchers to find and request datasets across the UK health data ecosystem – and DARE UK. Developed strategic partnerships with NHS Digital, Office of National Statistics, UKRI, academic institutions, and international bodies to ensure interoperability and secure data sharing across organisational boundaries. Championed Data Use Registers for transparency and accountability, establishing trust frameworks and data trustworthiness standards now foundational to the UK's national health data strategy.
- **COVID-19 National Response:** Led technology delivery underpinning the UK's COVID-19 health data analysis across all four nations, providing research and analysis reports directly to SAGE. Coordinated linked electronic health records covering 54 million people through the COVID-19 National Core Studies programme, demonstrating ability to mobilise data infrastructure rapidly at national scale.

EMBL - European Bioinformatics Institute

CAMBRIDGE, UNITED KINGDOM

Global Strategic Portfolio Manager - TSI, ELIXIR, EOSC-Life, EOSC-Hub & GA4GH Dec 17 – Jan 20

- **Major Programme & Data Infrastructure Leadership:** Led the €26M EOSC-Life and €33M EOSC-Hub European and international GA4GH & HDR-UK infrastructure programmes, managing complex multi-stakeholder budgets across 50+ partner institutions spanning genomic biobanks, research institutions, and national health data services. Contributed to the strategic management of EMBL-EBI's 400+

petabyte data estate – one of the world’s largest repositories of genomic and molecular data, underpinning research that contributed to the Nobel Prize-winning AlphaFold protein structure predictions. Co-authored EMBL-EBI’s **Hybrid Cloud Strategy** report, establishing frameworks for cost-effective, scalable cloud architecture for petabyte-scale genomic data.

- **Data Standards & FAIR Principles (June 2017 – November 2017):** Led strategic vision and technical roadmap for Human Data Communities in ELIXIR, defining standards for responsible sharing of sensitive human genomic and health data across European research infrastructures. Orchestrated €19M+ ELIXIR EXCELERATE initiatives, aligning technical platforms with scientific objectives. Pioneered applications of FAIR principles (Findable, Accessible, Interoperable, Reusable) in pharmaceutical and biomedical datasets, establishing data quality and interoperability standards adopted across the research community.

University of Sheffield

SHEFFIELD, UNITED KINGDOM

CTO - Translational Technology Officer - CISTIB, Scientific Workflows Coordinator - VPH-Share, Insigneo Institute
Jan '10 – Mar '17

- **Budget & Programme Management:** Architectural design and delivery of €18M **VPH-DARE@IT** hybrid cloud platform, managing budget across multiple funding streams and partner institutions. Development and operation of €13.4M **VPH-Share** cloud platform, demonstrating financial stewardship of complex distributed infrastructure. Coordinated 25+ research projects with combined budgets exceeding €30M, ensuring deliverables aligned with funder requirements.

- **Technical Leadership & Research Platforms:** Managed GIMIAS biomedical image processing workbench and developed research-as-a-service capabilities for 70+ multi-modal clinical datasets, providing researchers with intuitive access to complex biomedical data. Fostered interdisciplinary collaboration across academia, industry, NHS clinical partners, and healthcare organisations, building early expertise in translating clinical data into research-ready resources.

Please refer to my [LinkedIn](#) profile for a more complete list of work experiences along with recommendations.

Education

University of Milan-Bicocca

MILAN, ITALY

Executive MBA - Management of Research Infrastructures

2018 – 2020

Strategic, financial, and operational competencies for leadership of international research infrastructures. Focus on business models, sustainability planning, and cross-border governance

University of Sheffield

SHEFFIELD, UNITED KINGDOM

Doctor of Philosophy in Computational Systems Biology & Pure Mathematics

2005 – 2010

Hybrid, Hierarchical Models of Cardiac Cells and Tissues. ORS & Departmental scholarship, WUN Travel grant. SET Poster at House of Commons. Co-developed **FLAME** and **CHASTE** Frameworks.

Master of Science (Engineering) in Advanced Software Engineering

2004 – 2005

Distinction. Recipient of the Fretzwell-Downing Award for the Best Dissertation.

Pondicherry University

PONDICHERRY, INDIA

Bachelor of Technology in Information Technology

2000 – 2004

First class (86%) with Honors - Double Dissertation

Fellowships, Grants, Awards, Certifications, Board Memberships & Honours

Visiting Professor - **University of Sheffield**

2026 –

GA4GH AI Work Stream Co-lead - **Global Alliance for Genomics and Health**

2026 –

Scientific and Industry Board Member - Genomic Data Infrastructure (1+NG) project & MONARCH Initiative – advising on governance and access frameworks for large-scale population genomic datasets
2024 –

Member of the UKRI Infrastructure Advisory Committee – shaping national strategy for research data infrastructure including health data platforms
2024 –

Sponsor - Digital Regulation Cooperation Forum, BSI/ISO ART/1 Committee Member

2022 – 2024

Fellow of the British Computer Society & Member of the Association of Computing Machinery

2021 –

FitSM - Advanced Service Planning and Design & Advanced Service Operation and Control

2019 –

Azure for Research Grant - \$40,000, AWS in Education - £16,000, WANDisco - \$60,000

2013 - 2016

Lifetime Fellow of Software Sustainability Institute & Member of the SocRSE

2015 –

Worldwide Universities Network Travel Grant & SET for Britain Poster at House of Commons

2006

Fretzwell-Downing Award for Best MSc Dissertation, ORS & Departmental PhD Scholarship

2005

Selected Publications & Reports

- M Akhtar *et al.*, S Varma, *Croissant: A Metadata Format for ML-Ready Datasets*, NeurIPS, 2024
- A Pearson *et al.*, S Varma, *AI & Data Protection Risk Toolkit*, Information Commissioner's Office, 2022
- S Varma, T Hubbard, D Seymour, *Building Trusted Research Environments - Principles and Best Practices; Towards TRE ecosystems*, Dec 2021
- H L Rhem *et al.*, S Varma, E Birney, *GA4GH: International policies and standards for data sharing across genomic research and healthcare*, Cell Genomics, 2021
- J Lawson, [...], S Varma, M Courtot, *The Data Use Ontology to streamline responsible access to human biomedical datasets*, Cell Genomics, 2021
- Keynote Speaker at joint **European Commission & WHO** session on FAIR Data for future pandemic preparedness, 2021 & International Data Week 2022
- A Wood *et al.*, CVD-COVID-UK Consortium, *Linked electronic health records for research on a nationwide cohort of more than 54 million people in England*, BMJ, 2021
- G Saunders *et al.*, S Varma, S Scollen, N Blomberg, *Leveraging European infrastructures to access one million human genomes by 2022*, Nature Reviews Genetics, 2019
- M Fiume *et al.*, S Varma, S Scollen, *Federated discovery and sharing of genomic data using Beacons*, Nature Biotechnology, 2019

Please refer to my [ORCID:0000-0003-1687-2754](https://orcid.org/0000-0003-1687-2754) profile for a more complete list of scientific research outputs

Open Source Software, International Standards & Policies Contributions

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| • 2024-: Synapse.org, Synapse MCP Server | • 2024-: Synapse Knowledge Portals |
| • 2023-2023: How do we ensure fairness in AI? | • 2022-2023: Guidance on AI & Data Protection |
| • 2021-2022: GA4GH Computational Cohort Representation | • 2020-2022: HDR Innovation Gateway, Github |
| • 2020-2022: HDR UK Schemata | • 2020-2022: HDR UK Datasets, Papers/Preprints |
| • 2020-2022: HDR UK Phenotype Library | • 2020-2022: HDR UK Clinical Trials |
| • 2018-2020: GA4GH TRS, WES, TES & DRS Standards | • 2018-2020: EMBL-EBI ECP & AAP |
| • 2018-2020: ELIXIR TESK & WES-ELIXIR | • 2018-2020: EMBL-EBI RDSDS |
| • 2017-2020: GA4GH Beacon Network | • 2017-2020: Bioschemas Specification |
| • 2015-2017: VPH-DARE@IT Platform | • 2015-2017: GIMIAS Biomedical Workbench |
| • 2014-2015: VPH-HF Hypermodelling Framework | • 2011-2017: VPH-Share Cloud Platform |
| • 2006-2009: CHASTE | • 2004-2010: FLAME |